SEMINAR REPORT

on

Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method

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ABSTRACT

K-band Doppler wind profile radars are widely used to estimate atmospheric wind velocity and direction; however, weak atmospheric echoes are often corrupted by noise, clutter, and multiple Doppler spectral peaks under low signal-to-noise ratio (SNR) conditions. This paper proposes a Mutual Convolution Cost Function (MCCF) algorithm to improve the robustness of wind-profile estimation. The method preprocesses the radar echoes, adaptively constructs a space-time Doppler window based on wind continuity and wind-shear constraints, applies Mutual Convolution with multiple delay times to suppress uncorrelated noise while preserving coherent atmospheric echoes, and employs an improved Mutual Convolution Complex Least Mean Square (MCLMS) adaptive filter for further clutter suppression. Candidate Doppler peaks are then evaluated using a cost function that jointly considers spectral power and velocity continuity to identify the most probable atmospheric echo. Experimental evaluation using real VORTRAD K-band radar data demonstrates that the proposed method improves echo SNR by approximately 2–4 dB, significantly reduces false spectral peaks, produces smoother and more physically consistent wind profiles, and achieves correlation coefficients of approximately 0.95 with collocated LiDAR measurements, outperforming the AME, MPCF, and IME methods under low-SNR and multiple-peak conditions.

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List of Abbreviations

AME:	Adaptive Moment Estimation
CA:	Coherent Accumulation
CLMS:	Complex Least Mean Square
DC:	Direct Current
FFT:	Fast Fourier Transform
IME:	Improved Moment Estimation
MCCF:	Multiple Convoltion Cost Function
MCLMS:	Mutual Convolution and Complex Least Mean Square
MPCF:	Multiparameter Cost Function
PRI:	Pulse Repetition Interval
SNR:	Signal-to-Noise Ratio
ST–W:	Space-Time Doppler Window

CHAPTER 1
INTRODUCTION

The research paper titled *"Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method"* proposes a new method to estimate wind profiles by combining several methods including MCCF. The paper primarily focuses K-Band radar and on its potential application of such as efficient wind energy generation as well as challenges posed by seasonal atmospheric variation in measurement of near surface (< 1 km) wind turbulence.

1.1 Problem Addressed

The primary problem is that radar echoes from atmospheric turbulence are often heavily corrupted by strong background noise, resulting in a low SNR. This issue worsens in colder environments where decreased air density further weakens the returning signal. Additionally, the circular detection area of the radar antenna (widening of beam in higher altitudes) causes the returning echoes to frequently contain multiple spectral peaks. This makes it difficult to isolate the true frequency offset associated with the actual wind target.

Traditional data processing methods attempt to extract the wind profile by analyzing coherent accumulation (CA) signals with a cost function. While these methods work well when the SNR is high, their accuracy deteriorates under low SNR conditions with multiple competing peaks. Under these constraints, traditional algorithms become highly susceptible to clutter and noise, often mistaking false peaks for the true wind signal, which degrades the overall accuracy of the radar.

1.2 Overview of Process

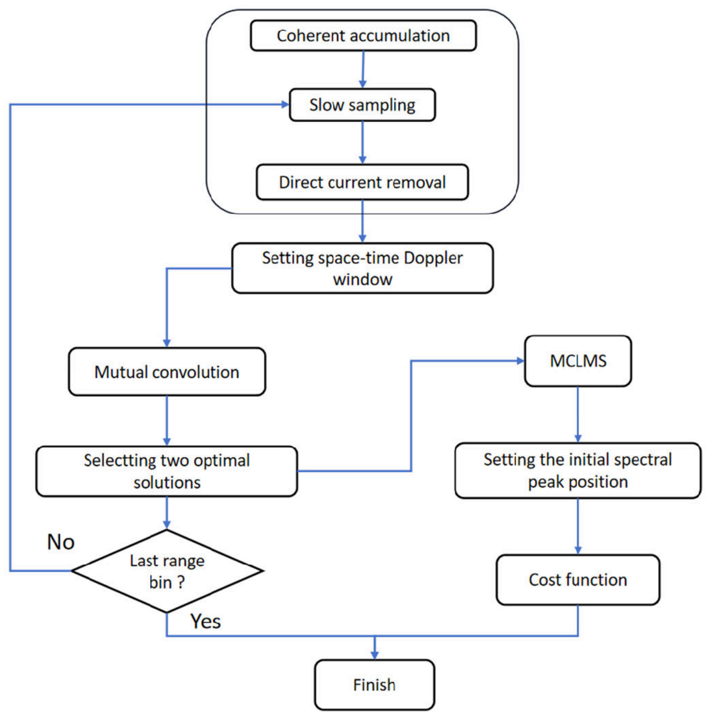
The paper describes a 5 step process to extract wind velocity:

1. **Signal Preprocessing**: Performs CA which boosts the steady wind signal and averages out random background noise. Resamples data into specific altitude slices (range bins) to track temporal changes in wind speeds at different heights. Finally atmospheric clutter and ground reflections are removed.
2. **Space-Time Doppler Window**: The system uses recent measurements to set a window in both spacial and temporal range. Any frequency peaks that fall far outside this window are ignored.

3. **Mutual Convolution**: The system takes an initial segment of radar signal at a height and compares it against slightly delayed segments of the same signal. It applies convolution with them and because random noise does not repeat exactly over time, it is canceled out. The stable wind signal persists across the segments and is amplified.
4. **Modified Complex Least Mean Square (MCLMS) Filter**: It minimizes the difference between the desired signal and the actual output, further smoothing the data and eliminating leftover clutter.
5. **Cost Function Evaluation**: The final step chooses the correct wind speed from any remaining signal peaks, by assigning penalty score to different aspects of the peaks and selecting the one with least penalty.

Finally the results are validates using data from a LIDAR operating in the same time period. The overall process is summarized in Figure 1.1

Figure 1.1: Overall flow of the method



1.3 Experiment Setup

The authors have utilized the K-band vortex radar (VORTRAD) as experiment equipment provided by Suzhou Dufeng Technology Company, which estimates wind profile by detecting Doppler frequency offset generated by atmospheric turbulence. The specification of VORTRAD system is given in Table 1.1. All experiments are conducted at Changxing Island, China.

The experiment setup is shown in Figure 1.2. The VORTRAD is equipped with a shield and two devices are placed 3 m apart. The distance resolution, velocity resolution, and measurement distance of the two devices are consistent, and the accuracy of this LIDAR under clear sky conditions reaches 99%. For comparing the effectiveness of proposed method, researchers have compared MCCF against CA, AME, MPCF, and IME using SNR and Correlation coefficient

Table 1.1: Specification of VORTRAD Radar

Number	Specification	Value
1	Frequency band	K-band
2	Operation mode	Single frequency pulse
3	Number of beams	4 (East, South, West, North)
4	Tilt Angle	15°
5	Pulse period	2us
6	Pulse width	≈ 133.3ns
7	No.of coherent accumulation	50
8	Slow sampling points	2048
9	FFT points	1024
10	Maximum range	300m
11	Window coefficient	2
12	Maximum radical velocity	30 m/s
13	Radial velocity resolution	0.06 m/s
14	Range resolution	20m

Figure 1.2: Experiment setup

CHAPTER 2
RESEARCH METHODOLOGY

This section introduces the data processing algorithm in detail and mainly addresses the issues of low SNR and multiple spectral peaks by the MCCF algorithm. The algorithm is divided into five parts, signal preprocessing, setting the space-time Doppler window, mutual convolution, MCLMS filter, and generating the cost function.

2.1 Signal Preprocessing

This phase prepares data for MCCF algorithm. In this phase, 3 operations are performed on data:

2.1.1 Coherent Accumulation (CA)

Here atmospheric turbulence is characterized as a weak target. The model assumes that turbulence moves at a uniform velocity over a short period. Hence its echo is akin to a stable periodic signal. Conversely, noise exhibits instability and lacks periodicity. The CA adds and averages short segments of radar signals thereby reducing noise strength and amplifying turbulence echo. Throughout the paper 2-D matrix generated by the result after N-order CA is represented as:

$$X = \left[x_{c1}^T, x_{c2}^T, ..., x_{c(N-1)}^T, x_{cN}^T\right]$$

Where x_{ci} is result of i^{th} CA. (Hence the column vector of the X represents the result of one accumulation)

2.1.2 Slow Sampling

To detect the motion state of turbulence at different altitudes, we need to extract data from a specific range bin. When a pulse is transmitted, the radar receives echoes from all range bins (heights). After one Pulse Repetition Interval (PRI), it transmits the next pulse and repeats the process. For a single range gate (height), the radar stores one complex echo sample from every transmitted pulse. This sequence is called *slow-time signal*. To obtain this, a row vector from X is used. This process is called *slow sampling* in the distance dimension. The slow sampling frequency must satisfy the Nyquist criterion in order to avoid blurring the maximum turbulence velocity, and the slow sampling frequency $f_{d\min}$ can be given as

$$f_{d\min} = \frac{4v_{\max}}{\lambda}$$

where $v_{\max}$ represents the maximum velocity range and λ represents the work wavelength of the radar

2.1.3 DC and Clutter Removal

Hardware components and ground reflections create direct current (DC) interference, which shows up as clutter near the zero-frequency point. In case of VORTRAD which adopts a zero-intermediate-frequency RF transceiver architecture, produces signal containing a large dc power, and ground clutter with a small frequency offset. The paper doesn't mention a particular clutter removal method.

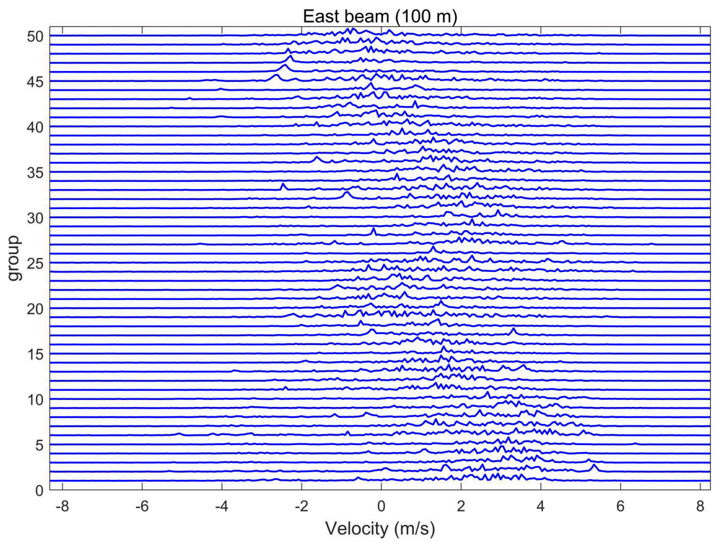
2.2 Setting the Space-Time Doppler Window

In this stage, the search window for peaks is reduced by using 2 property of wind velocity:

- Turbulent motion state between adjacent altitudes continuously changes and there is a maximum wind shear associated with a difference in altitude (typically less than 0.5 m/s in the range of 10 m and is less than 1 m/s in the range of 20 m).
- Wind velocity exhibits continuity over multiple pulse same altitude as shown in Figure 2.1

Therefore, when setting this search window (doppler window), it is essential to consider both spatial and temporal aspects. The size of doppler window.

Figure 2.1: Wind velocity over 50 consecutive groups at 100 m. (provided by researchers)



The researchers have quantitatively modeled the window size using frequency offset corresponding to wind shear (d_f) and "window coefficient related to altitude interval" (α). First, considering space continuity, assuming that the spectrum estimation result of the previous range bin is F_0 , the space Doppler window can be represented as

$$W_1 = \left(F_0 - \alpha d_f, F_0 + \alpha d_f\right)$$

Where d_f is given by:

$$d_f = \frac{V_d}{d_v} = \frac{V_d}{f_{d\min}\lambda/L}$$

Where V_d = maximum value of wind shear and L = number of points in FFT.

Now, considering time continuity, and assuming the previous spectrum estimation result at the same range bin is T_0, the time Doppler window can be represented as

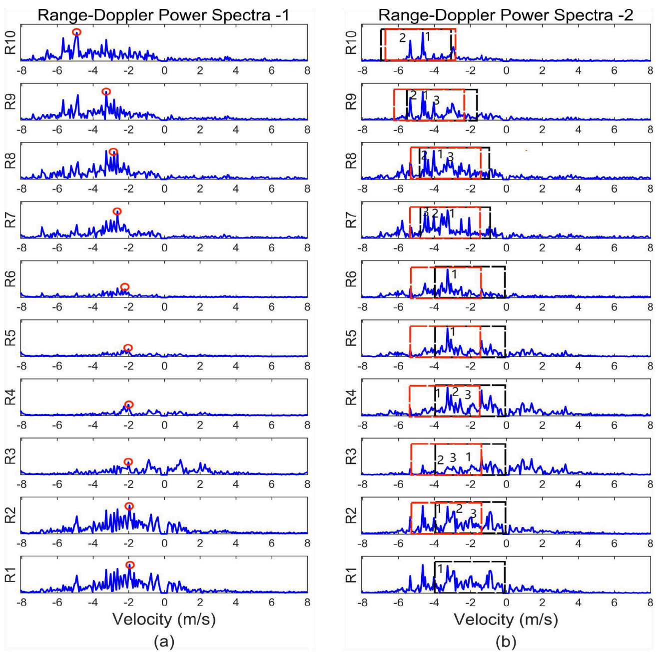
$$W_1 = \left(T_0 - \alpha d_f, T_0 + \alpha d_f\right)$$

The window used for peak estimation is intersection of these 2 windows, the researchers have termed this window as *space-time Doppler window*, expressed as:

$$\mathrm{ST-W} = \begin{cases} W_1 \bigcap W_2 & \text{if } W_1 \bigcap W_2 \geq \alpha d_f \\ W_1 & \text{else} \end{cases}$$

An example case of ST–W and peak estimation is illustrated in Figure 2.2

Figure 2.2: (a) spectra peak selection highlighted in red circles (b) Space (red) and Time (black) doppler windows marked



2.3 Mutual Convolution

To minimize the error rate of spectral estimation, efforts should be made to attenuate noise and retain significant spectral peaks before identifying candidate spectral peaks. The main idea is to exploit the temporal correlation of atmospheric echoes while suppressing random noise which is highly uncorrelated. Suppose successive pulses observe the same moving air parcel. The received signal changes slowly. Similarly, its phase also changes smoothly. Therefore, successive slow-time samples are strongly correlated.

Because of this difference, the researchers used mutual convolution to measure the similarity between delayed versions of the slow-time signal.

The method is described below:

First, take a row vector $P = (p_1, p_2, ..., p_N)$ of one range bin from the preprocessing data X. Then, take out vectors with equal length from P to form a new row vector. Here the researchers used half the size of the orignal vector as new vector. Let these constructed vectors be:

$$V_{r1} = \left(p_1, p_2, ..., p_{N/2}\right)$$

The half length for sequences allows several overlapping signal segments to be created. The paper defines number of delay points I as:

$$I = \frac{N}{2(n-1)}$$

where

- N = total length of vector P
- n = desired number of extracted vectors

From this a new matrix can be given as:

$$V = [V_{r1}, V_{r1},, ..., V_{rn},]$$

Now mutual convolution is performed:

$$\begin{aligned} V_{C1} &= V_{r2} * V_{r1}^{\mathrm{HR}} \\ V_{C2} &= V_{r3} * V_{r1}^{\mathrm{HR}} \\ &\vdots \\ V_{C(n-1)} &= V_{rn} * V_{r1}^{\mathrm{HR}} \end{aligned}$$

After the above calculation, the number of convolution results is $n-1$ and the length of each calculation result becomes $N-1$. To maintain the same time complexity and velocity resolution of calculation, a new matrix can be generated by extracting data from the middle of the vector, with the number being $N/2$, given as:

$$[V_C'] = \begin{bmatrix} V_{C1}' \\ V_{C2}' \\ \vdots \\ V_{C(n-1)}' \end{bmatrix} = \begin{bmatrix} V_{C1}\left(\frac{N}{4} : \frac{3N}{4}\right) \\ V_{C2}\left(\frac{N}{4} : \frac{3N}{4}\right) \\ \vdots \\ V_{C(n-1)}\left(\frac{N}{4} : \frac{3N}{4}\right) \end{bmatrix}$$

Then maximum of power spectra of each row is computed:

$$V_{cf}' = \left[\max\left\{\left|\mathcal{F}\left([V_c']\right)\right|\right\}\right]$$

where $\mathcal{F}$ represents Fourier transform. The maximum of V_{cf}' corresponds to some optimial solution X_r (let that be V_{Ck}'). In practice the researchers choose to keep adjacent readings ($V_{C(k-1)}'$ and $V_{C(k+1)}'$) which are kept as candidate peaks or suboptimal soluions X_r'.

2.4 MCLMS

MCLMS is a adaptive filter. Its purpose is to further suppress residual clutter and random noise after the Mutual Convolution stage has identified the most coherent signal segment. The algorithm uses the optimal convolution result X_r and the adjacent suboptimal convolution result X_r' as the input and desired signals of a CLMS adaptive filter. The underlying assumption is that these two signals contain the same atmospheric echo because they originate from adjacent delay windows, while their noise and clutter components are largely uncorrelated. By adapting the filter to transform X_r into X_r', the filter reinforces the common atmospheric component and suppresses the components that differ between the two signals. Unlike a conventional adaptive filter, where adaptation and filtering occur simultaneously, the proposed MCLMS algorithm separates these operations into two stages. During the first stage, the filter is used only for training – the adaptive filter coefficients are continuously updated until they converge. After convergence, the learned coefficients are copied and reused in a second filtering pass to generate the final filtered output. The specific calculation process is as follows. The optimal solution is represented as an $\frac{N}{2} - N/2$ - dimensional row vector and the length of the coefficient filter is denoted as k. First a normalization factor is calculated.

$$u = \frac{1}{\sum_{j=1}^{N/2} |X_r(j)|^2}$$

This balances how coefficients are adjusted even when one input is much larger in comparison to others. The adaptive filter has a length of k coefficients. For every sample position j, an input vector X is constructed from the samples of X_r. If there are insufficient previous samples at the beginning of the sequence, zeros are inserted.

$$X = \begin{cases} \text{zeros}(1,\ k-j,\ X_r(1:j)) & ,\ j < k \\ X_r(j-k+1:j) & ,\ j < k \end{cases} \qquad j \in \{1,\ 2,\ ..., N/2\}$$

This effectively forms a sliding window over the signal, allowing the adaptive filter to operate as a finite impulse response (FIR) filter. The filter coefficients are initially set to zero,

$$W_1 = 0$$

For each input vector, the filter output is calculated using

$$Y_j = W_1 X^T$$

which is simply the dot product between the current coefficient vector and the input vector. The difference between the desired signal X_r' and the filter output produces the estimation error,

$$e_j = X'_{r(j)} - Y_j$$

The coefficients are then updated using

$$W_1' = 0.98W_1 + ue_j X^H$$

The factor 0.98 is a leakage factor that slightly reduces the magnitude of the coefficients during each iteration, preventing them from growing excessively and improving numerical stability

The coefficient update process is repeated for every sample in the signal until all data have been processed. At this stage, the coefficients have converged to values that best represent the common characteristics shared by X_r and X_r'. finally the second filter coefficient is W_1' itself. In the second pass, the algorithm repeats the filtering operation using the fixed coefficient vector W_2 but without further adaptation. Since the coefficients have already converged, the filter produces a stable output Y_r, which contains a stronger atmospheric echo and significantly reduced noise and clutter. This filtered signal is subsequently used by the MCCF stage for reliable Doppler peak identification and wind profile estimation.

2.5 Generating the Cost Function

After the Mutual Convolution and MCLMS stages, the radar signal has been significantly cleaned, and a Doppler spectrum has been obtained for every range bin. However multiple spectral peaks may still exist within a single Doppler spectrum. In conventional spectral estimation methods, it is generally assumed that the first range bin contains only one dominant spectral peak, which is selected as the atmospheric echo. The researchers point out that this assumption is not valid for the VORTRAD and other wind profilers because strong ground clutter, interference, and other unwanted echoes often produce multiple prominent peaks in the first range bin. Consequently, selecting the strongest peak directly can result in incorrect wind estimation. To overcome this problem, the paper introduces the Mutual Convolution Cost Function (MCCF), which evaluates candidate spectral peaks using both their spectral power and their physical consistency with neighbouring range bins.

first step is to calculate the average value corresponding to the power spectrum of the first range bin, it can be given as:

$$M_1 = \frac{\sum_{i=1}^{n-1} \mathcal{F}(V'_{Ci})}{n-1}$$

The resulting average spectrum provides a more stable estimate of the true Doppler spectrum in the first range bin. The algorithm then identifies the strongest peak in this averaged spectrum,

$$M_{1,1} = \max|M_1|$$

which serves as the initial reference for subsequent peak tracking through higher range bins. For every range bin, the Doppler spectrum is then computed from the MCLMS filtered signal, $M_r = |\mathcal{F}(Y_r)|$

To reduce the possibility of selecting insignificant noise peaks, the algorithm first sorts all spectral peaks according to their power and retains at most three candidate peaks within each ST–W. Furthermore, a candidate peak is considered only if its power satisfies

$$M_{r,j} \geq 0.4 M_{r\max}$$

Where $M_{r\max}$ is maximum spectral peak within the ST–W and $M_{r,j}$ represents the power value corresponding to the jth spectral peak of the rth range bin. This threshold removes weak noise peaks. The algorithm then evaluates how well each candidate peak agrees with the wind velocity estimated in the previous range bin. Atmospheric winds normally vary smoothly with altitude, and therefore the Doppler velocity should also change gradually between adjacent range bins. To quantify this behaviour, the paper defines a velocity cost

$$V_{r+1,j} = 1 - \left| \frac{(M_{r+1} - M_r)d_v}{\alpha V_d} \right|$$

where

- d_v = velocity resolution
- V_d = maximum wind shear
- α = window coefficient,
- $V_{r+1,j}$ = velocity cost value corresponding to the jth

spectral peak of the $(r+1)$th range bin.

Finally, the algorithm combines two independent criteria to evaluate each candidate peak – velocity continuity (represented by velocity cost $V_{r+1,j}$) and spectral power of the candidate peak ($M_{r+1,j}$). These two constraints are combined through the MCCF cost function:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \qquad \text{and } w_1 + w_2 = 1$$

where w_1 and w_2 are weighting factors controlling the relative importance of velocity continuity and spectral power. This weighted cost function ($F_{r+1,j}$) forces selection of candidate peak that simultaneously possess sufficient spectral power and reasonable continuation of the wind profile. The peak with the highest cost value is selected as the atmospheric Doppler peak for that range bin, and the process is repeated for all subsequent range bins until the complete wind profile has been estimated.

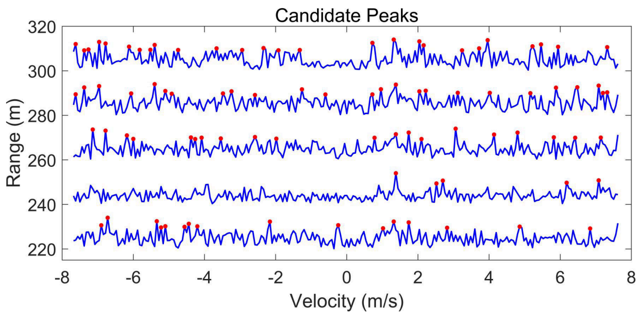
CHAPTER 3
RESULTS

The authors have evaluated the algorithm in four ways: Delay time for mutual convolution, effectiveness in reducing clutter and improving SNR, comparing wind profile estimate with existing algorithm, validating the estimated wind velocities against LiDAR using correlation coefficient. Each experiment methods gives a different aspect of the proposed algorithm.

3.1 Performance Evaluation at Different Delay Times

This experiment investigates how the delay time used in the Mutual Convolution stage affects clutter suppression. The algorithm correlates the original signal with delayed versions of itself. If the delay is too short, the clutter and noise remain highly correlated and are not sufficiently suppressed. If the delay is too large, even the atmospheric echo begins to decorrelate, reducing the effectiveness of the filtering.

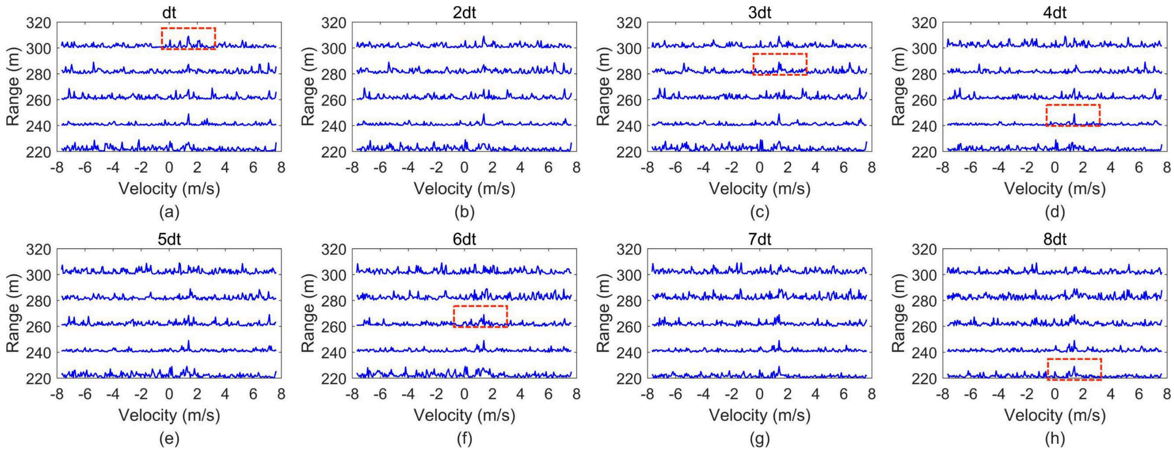
Figure 3.1: Range-Doppler power spectra after preprocessing



The researchers begin by showing the preprocessed Doppler spectra in Figure 3.1. These spectra correspond to echoes between 220 m and 300 m altitude. The red dots represent all spectral peaks that satisfy the candidate selection criteria. One important observation is that only the 240 m range bin exhibits a clearly distinguishable atmospheric peak, while the remaining range bins contain numerous peaks of nearly equal amplitude. This illustrates the primary challenge addressed by the paper: under low SNR conditions, simply selecting the strongest peak can easily lead to an incorrect wind estimate because many clutter and noise peaks have comparable power.

To determine the best delay, the researchers repeat the mutual convolution process using eight different delay values, ranging from d_t to $8d_t$. The filtering results are presented in Figure 3.2. Each subfigure shows the Doppler spectra obtained using a different delay. From this we can conclude that the optimal delay is not same for every altitude.

Figure 3.2: Effect of delay in optimal solutions (marked by red dotted box) of different range bin after mutual convolution



To further analyze this behavior, the authors process 2500 radar frames collected on January 11, 2023. The atmosphere is divided into eight 40 m range intervals, and for every frame they record which delay produces the strongest mutual convolution result. These statistics are summarized in Table 3.1. Although different delays occasionally become optimal, the smallest delay d_t is selected far more frequently than the others across all range bins. The frequency of selecting larger delays decreases steadily. Based on these statistics, the authors conclude that while the optimal delay varies with atmospheric conditions, shorter delays generally provide the most reliable filtering performance and excessively large delays should be avoided because they reduce signal correlation.

Table 3.1: Statistics of optimal delay time for mutual convolution

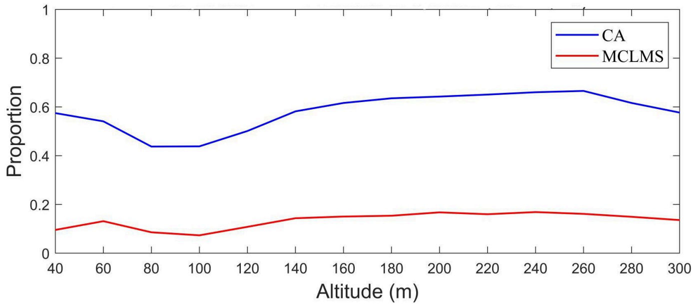
Delay	RI	R2	R3	R4	R5	R6	R7	R8
d_t	1543	1503	1568	1471	1486	1488	1498	1568
$2d_t$	444	404	430	420	398	397	406	403
$3d_t$	179	203	195	224	214	218	223	196
$4d_t$	128	129	91	119	114	127	122	126
$5d_t$	70	72	68	73	85	80	78	70
$6d_t$	39	63	58	69	75	71	66	35
$7d_t$	42	60	38	63	59	51	48	47
$8d_t$	55	66	52	61	69	68	59	55

3.2 Peak Number and SNR

Here researchers have compared number of candidate spectral peaks and the SNR obtained from CA and proposed MCLMS filters. Ideally, each Doppler spectrum should contain only one dominant atmospheric peak. If many peaks remain, the peak-selection algorithm becomes much more likely to choose an incorrect target. Using 5000 radar data sets, the researchers compare the conventional CA with the proposed MCLMS filter. The results are shown in Figure 3.3, which plots the proportion of spectra containing more than six candidate peaks. Across every altitude, the CA produces a high percentage of multi-peak spectra, typically between 45% and 70%. After applying the MCLMS filter, this proportion decreases dramatically to approximately 10–17%. This demonstrates that the adaptive filtering stage effectively suppresses clutter peaks while preserving the dominant atmospheric echo. The observations were recorded in following conditions:

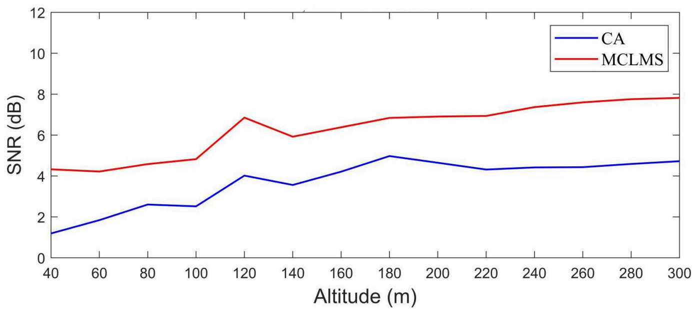
- **Time**: 6pm – 8pm (Dec 17, 2022)
- **Temperature**: −2 ~ 9 °C
- **Wind speed**: 4 ~ 8 m/s
- **Wind direction**: southeast

Figure 3.3: Proportion of the number exceeding 6 of candidate spectral peaks after applying the CA and the MCLMS filter respectively



The second performance metric is the SNR improvement, presented in Figure 3.4. Before filtering, the average SNR across all range bins remains below 5 dB, indicating very weak atmospheric echoes. After MCLMS filtering, the SNR increases by approximately 2–4 dB at every altitude. The higher SNR produces cleaner Doppler spectra and improves the reliability of subsequent peak selection.

Figure 3.4: SNR after applying the CA and the MCLMS filter respectively



3.3 Wind Profile Estimation Results

The researchers next evaluated the complete MCCF algorithm by comparing its wind estimates with three existing spectral estimation methods: AME, MPCF, and IME.

Figure 3.5 and Figure 3.6 present two radar observations recorded only one second apart. In Figure (a) each case shows the original Doppler spectra together with the wind profiles estimated by all four algorithms, while Figure (b) shows the spectra after MCLMS filtering together with the final MCCF estimates.

The researchers have observed that the AME and IME methods sometimes select clutter peaks in the lower range bins. Since subsequent peak selection depends on previous estimates, these initial errors propagate upward, producing unstable wind profiles with large deviations. The MPCF method performs somewhat better but still exhibits considerable fluctuations, particularly above 100 m, because its candidate selection region is relatively broad. In contrast, the MCCF algorithm generates a much smoother and more continuous wind profile. The selected Doppler peaks follow a physically realistic gradual change in atmospheric wind velocity with altitude. Even though the two observations are separated by only one second, the other algorithms produce noticeably different wind profiles, whereas MCCF maintains nearly identical estimates, demonstrating superior temporal stability.

Figure 3.5: Comparison of four methods based on the data recorded on December 18, 2022 (15:19:32)

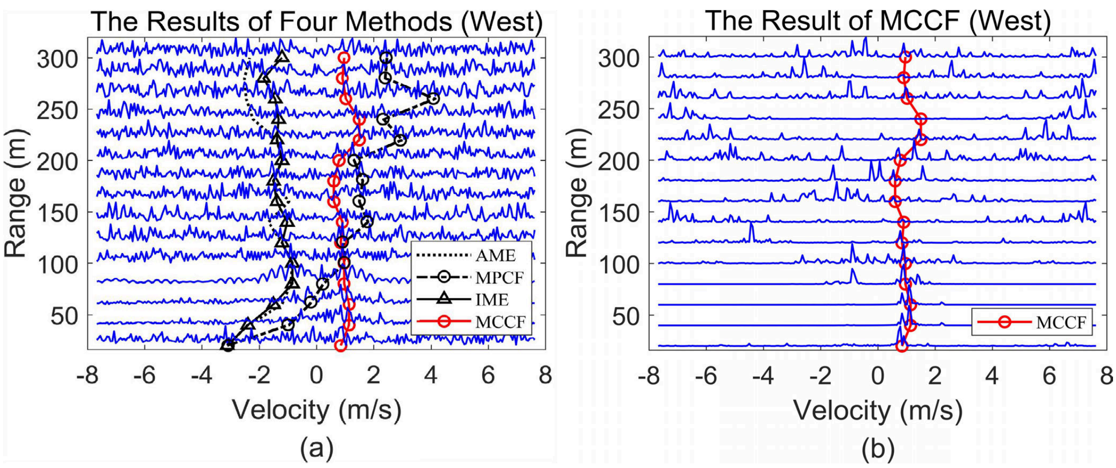
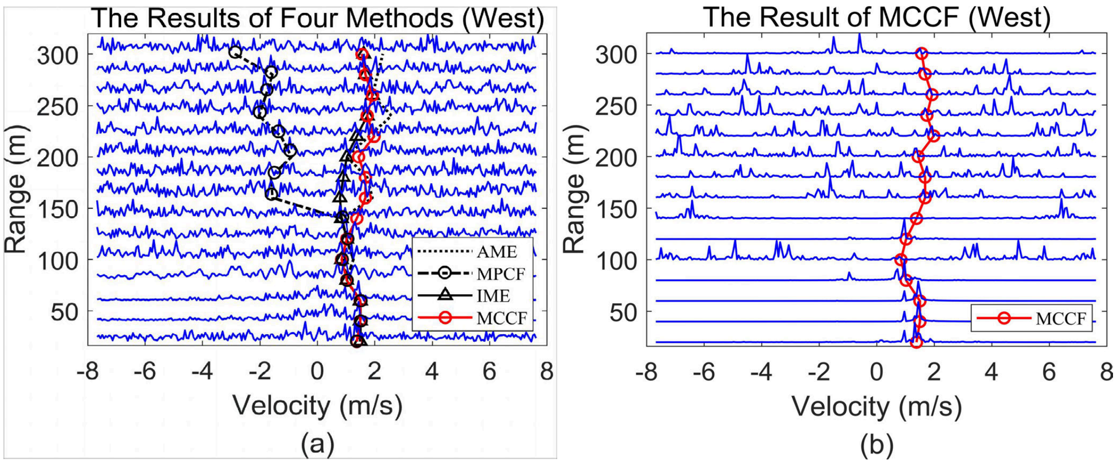


Figure 3.6: Comparison of four methods based on the data recorded on December 18, 2022 (15:19:33)



3.4 Scatter Plot and Correlation with LiDAR Observation

The final experiment validates the estimated wind velocities using an independent reference instrument – LiDAR. During March 3 to March 7, 2023, the VORTRAD radar and nearby LiDAR measured horizontal wind velocity. Because LiDAR generally has highest accuracy among all instruments, it provides a suitable reference for evaluating the radar algorithms.

For each method, the researchers compared 2500 pairs of radar and LiDAR measurements at an altitude of 200 m. The scatter plots shown in Figure 3.7 display radar-estimated velocity on the vertical axis and LiDAR velocity on the horizontal axis. The diagonal line represents perfect agreement between the two instruments.

The scatter plots clearly show that the MCCF estimates are much more tightly clustered around the diagonal than those produced by AME, MPCF, or IME. This indicates that the proposed algorithm produces wind velocity estimates much closer to the LiDAR measurements. The improvement is quantified using the correlation coefficient (CC), summarized in Table 3.2. Across all heights from 40 m to 280 m, the MCCF method consistently achieves correlation coefficients between 0.9576 and 0.9651, whereas the IME method reaches approximately 0.85–0.88, MPCF remains around 0.82–0.84, and AME performs worst, typically below 0.84. Since a correlation coefficient closer to unity indicates stronger agreement between two independent measurements, these results shows that the MCCF algorithm provides the most accurate wind velocity estimation among the four methods.

Figure 3.7: Scatter plot showing correlation among AME, MPCF, IME, MCCF methods against LiDAR. Color bars indicate data density

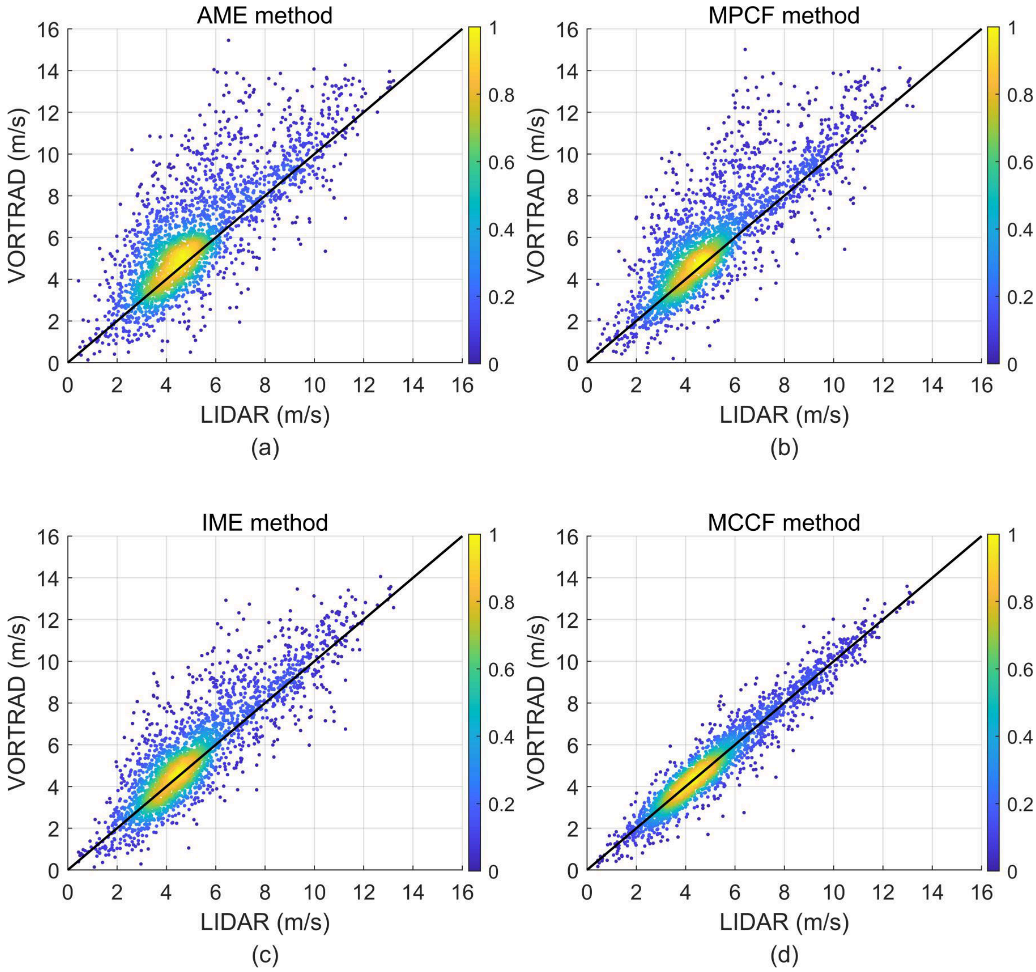


Table 3.2: Comparison of the correlation coefficient for VORTRAD and LiDAR data by using AME, MPCF, IME, MCCF

Height(m)	AME	MPCF	IME	MCCF
40	0.8417	0.8368	0.8369	0.9576
80	0.8319	0.8350	0.8461	0.9609
120	0.8320	0.8431	0.8695	0.9625
160	0.8137	0.8353	0.8761	0.9641
200	0.7908	0.8276	0.8764	0.9647
240	0.7830	0.8202	0.8738	0.9650
280	0.7837	0.8219	0.8744	0.9651

CHAPTER 4
CONCLUSION

Wind profile estimation using K-band Doppler radar is challenging under low SNR conditions because radar echoes are often contaminated by clutter, noise, and multiple Doppler peaks, making it difficult to identify the true atmospheric echo. To address this problem, a new method was proposed – Mutual Convolution Cost Function (MCCF) which combines the advantages of existing spectral estimation methods (AME, MPCF, and IME) with an improved CLMS filter. The proposed method first performs Mutual Convolution using multiple delayed versions of the radar signal to suppress uncorrelated noise while preserving coherent atmospheric echoes. An improved Mutual Convolution Complex Least Mean Square (MCLMS) filter is then applied to further reduce clutter and enhance the desired signal. Finally, candidate Doppler peaks are evaluated using a cost function that considers both spectral power and the physical continuity of wind velocity between adjacent range bins, enabling reliable selection of the atmospheric Doppler peak.

The proposed algorithm was evaluated using real atmospheric data collected by the VORTRAD K-band wind profiler. Experiments showed that the MCLMS filter reduced the number of false spectral peaks and improved the signal-to-noise ratio by approximately 2–4 dB. Compared with the AME, MPCF, and IME methods, the MCCF algorithm produced smoother and more continuous wind profiles while avoiding incorrect selection of clutter peaks. Validation against simultaneous LiDAR measurements demonstrated that MCCF achieved the highest correlation coefficient (approximately 0.96) across all tested altitudes, confirming its superior accuracy for wind profile estimation under low-SNR and multi-peak conditions. The authors note that the current method is validated only for clear-weather conditions and does not account for precipitation echoes such as rain or snow, leaving this as future work.